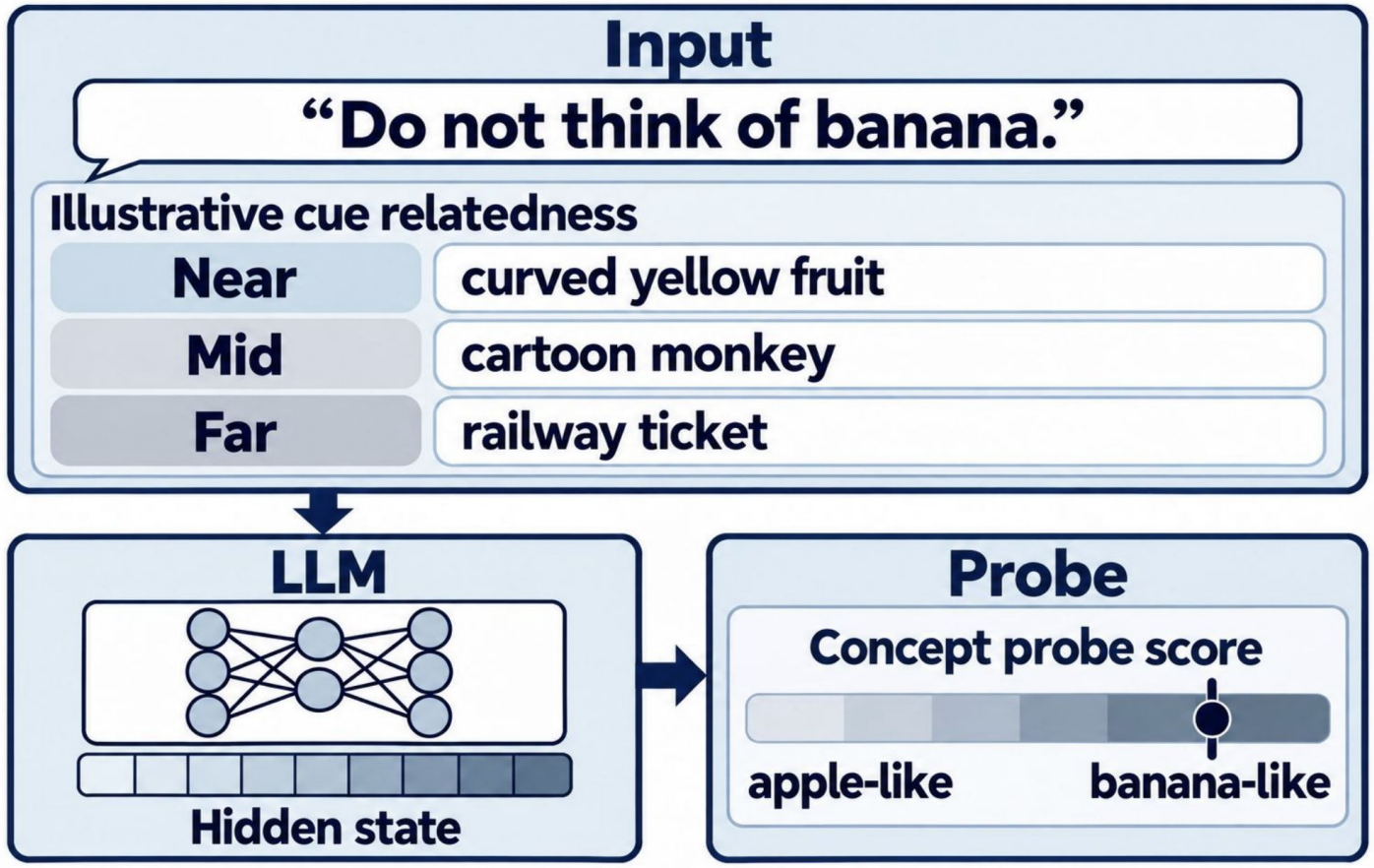
Semantic Distance as a Metric & Pilot Validation

意味的距離の測定と予備実験

Ryo Matsushita / 2025.11.04

Research Question

Does semantic distance between a cue and a target affect the LLM's internal representation of that target?

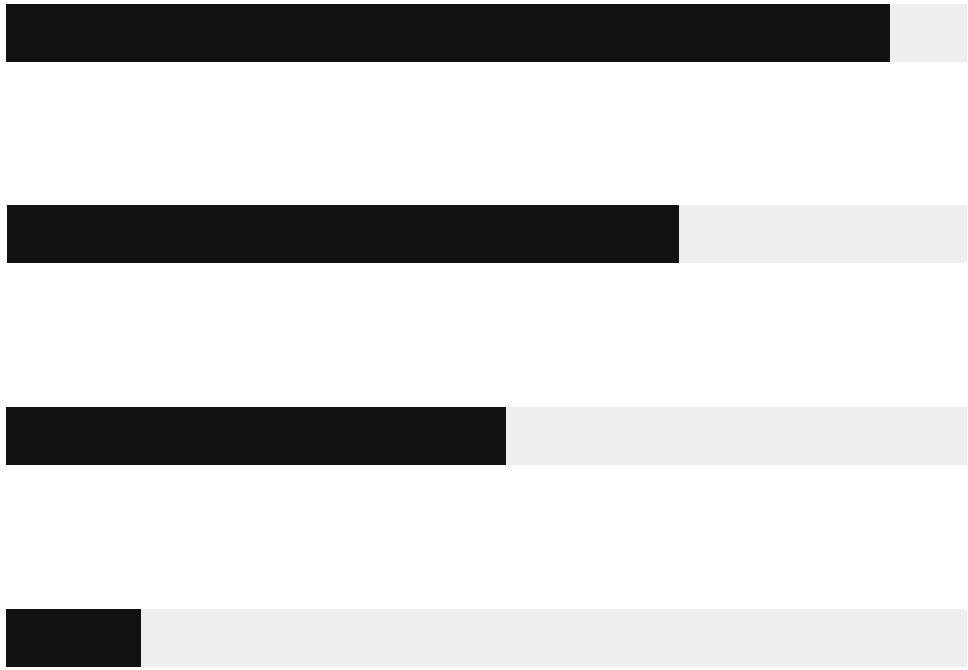


What is Semantic Distance?

A continuous measure of how conceptually close two words are.

Example (target = elephant)

Mammoth	Semantically similar	Near
Trunk	Feature-related	
Safari	Situationally related	
Algorithm	Weakly related	Far



Measuring with Cosine Distance

1. Encode

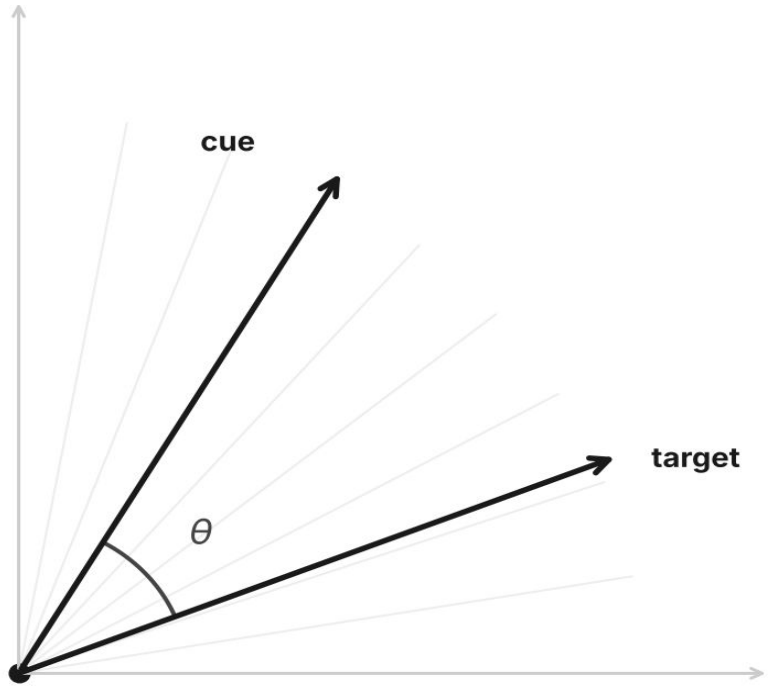
Target & Cue → embedding vectors

2. Cosine similarity

cos(target, cue) = dot product / (|a| × |b|)

3. Cosine distance

d = 1 − cos(X, Y) 0 = identical, 1 = unrelated



Selected Model

all-mpnet-base-v2

768-dim / English / Open-source

Why?

- Best quality general-purpose sentence similarity model
- Independent from Qwen — avoids measurement confound
- Open-source & reproducible

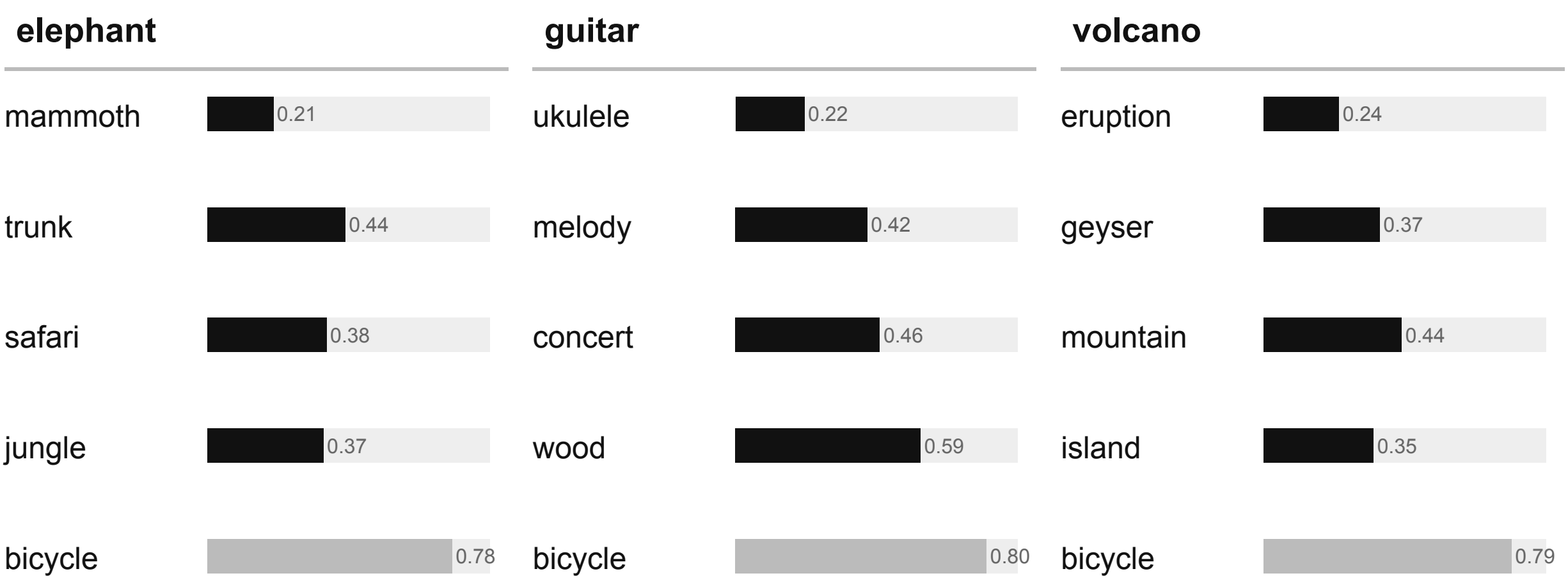
Note: Single-word validity and human-association alignment need verification.

Pilot Setup

Goal: Can we construct cues at graded semantic distances?

Target	← Near	Cue 2	Cue 3	Far →
elephant	mammoth	trunk	safari	bicycle
guitar	ukulele	melody	concert	bicycle
volcano	eruption	geyser	mountain	bicycle

Results: Cosine Distance Scores



Near cues scored lower as expected — but some rankings differ from intuition.

Key Findings

elephant: jungle < trunk (contextual > feature)

guitar: violin < ukulele (category > shape)

volcano: island < geyser (co-occurrence > feature)

Cosine distance works — but model distance ≠ human intuition.

Next Steps

1. Expand to 8 targets; select cues at 4 distance levels
2. Present cues to Qwen2.5-0.5B-Instruct
3. Measure internal rep. via Layer-wise Concept Probe
4. Compare across 4 instruction conditions

Questions for Discussion

Q1. Is our definition of semantic distance theoretically sound?

Q2. Is single-word embedding sufficient?

Q3. Is a 4-level distance gradient appropriate?

Q4. How do we evaluate changes in internal representation?

Probe Discrimination Performance

Objective

Can the probe correctly identify concepts with new cues?

Method

- Model: Qwen2.5-0.5B-Instruct
- 3 concept pairs
- Compare AUC: previous vs. new cues

Results

- AUC dropped for Banana & Elephant
- Piano maintained high performance
- Stability issues in 20-pair extension

Probe discrimination when cues change

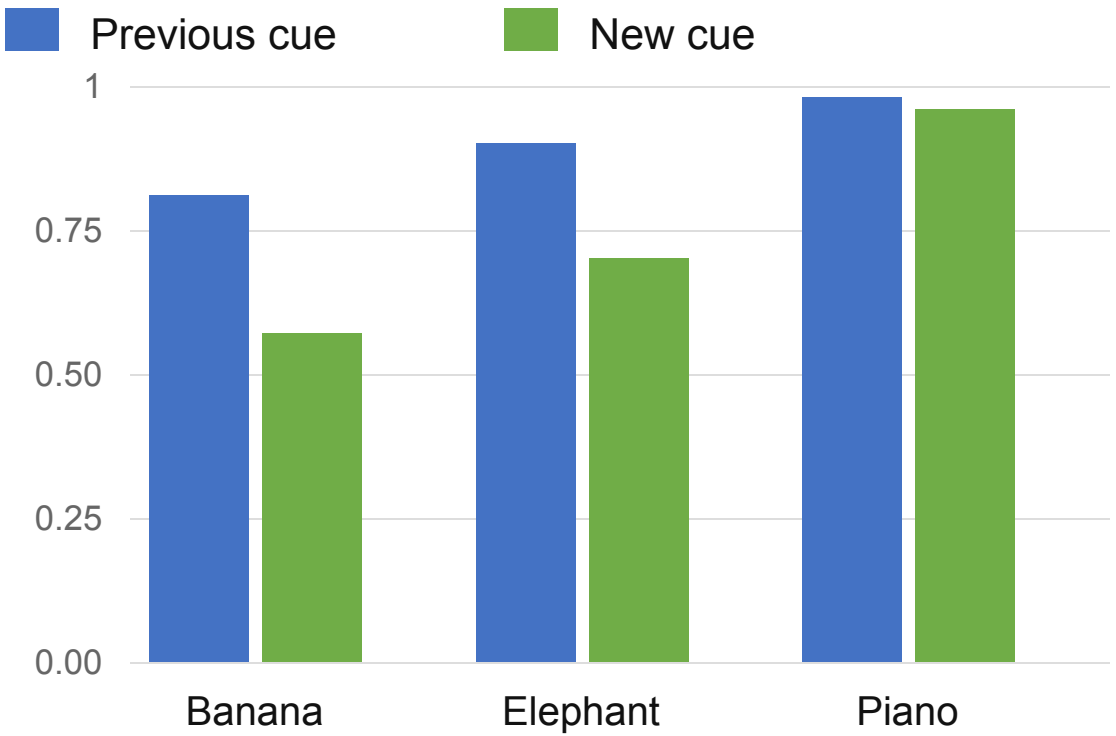


Fig.1 ROC-AUC comparison (avg. layers 22–23). Higher AUC = better discrimination.

Probe performance may vary depending on the specific cues used.

Semantic Distance & Internal Representation (Pilot)

Objective

Does semantic distance between cue and target affect internal representation?

Method

- Model: Qwen2.5-0.5B-Instruct
- 8 concept pairs
- Probe response vs. cosine distance under suppression

Results

- Suppression narrowed discrimination gap
- No clear pattern by semantic distance

Effect of distance on probe response (slope + 95% CI, 8 concept pairs)

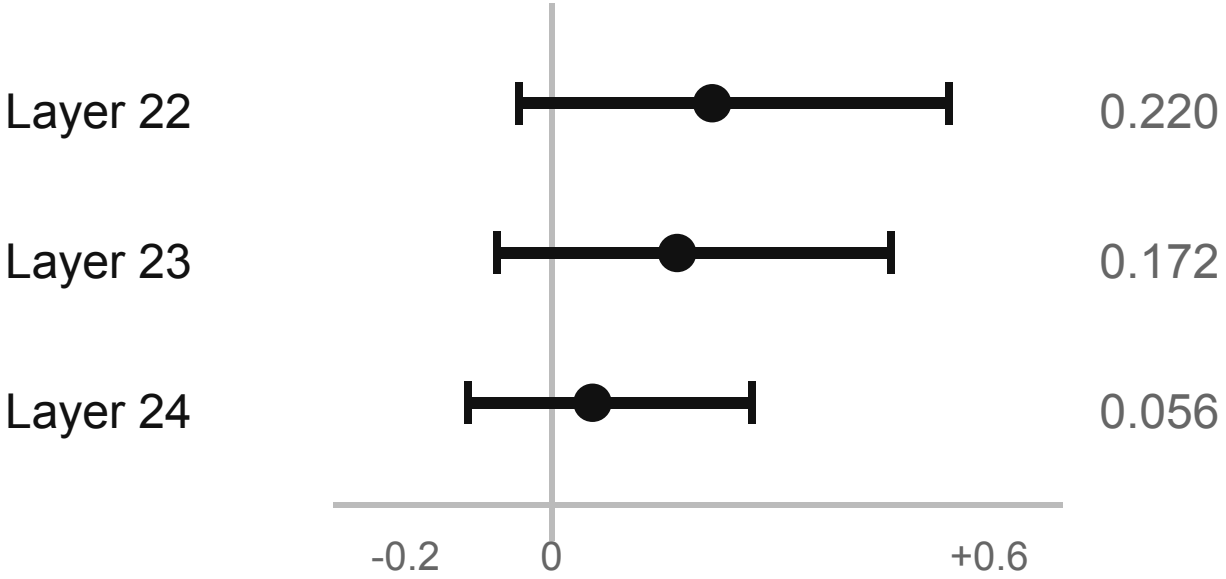


Fig.2 Slope of distance effect on probe response (exploratory). All 95% CIs include 0.
(Pilot data using MiniLM — to be revisited with all-mpnet-base-v2)

Suppression effect observed — but distance effect remains inconclusive.